

Comprehensive Phishing Email Investigation: Medisure Health Network Response and Findings

Detailed analysis of the phishing incident, scope, timeline, and remediation steps

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Organisation SOC Investigation report

Presenter



Executive Summary



Impersonate Email

Attackers mimicked a legitimate Microsoft account to solicit user credentials from targets.

Security Gateway

The email security gateway identified the malicious markers and quarantined the message.

Redirecting Email

Reply-to addresses were rerouted to an external Gmail account to facilitate data theft.

Malicious Tracking

Embedded tracking pixels monitored user interactions to verify active targets and activity.

Objectives

Project Objectives

01

Investigate a suspicious email claiming to be from **Microsoft**

02

Analyze **email headers** and authentication results

03

Identify **phishing indicators** and malicious infrastructure

04

Review **user activity** and network logs

05

Provide **recommendations** to improve security

Investigation Tools and Purpose

Key tools used to validate IP legitimacy, check authentication, and identify malicious infrastructure across domains and headers

- » Threat Intelligence Analysis Platform for aggregated threat context and correlation
- » AbuseIPDB to check IP reputation and geolocation
- » MxToolBox for header inspection and SPF/DKIM/DMARC authentication analysis
- » VirusTotal to scan URL and domain reputation across engines
- » Purpose: validate IP legitimacy, confirm SPF/DKIM/DMARC, and identify malicious infrastructure

Incident Overview

01

Email impersonated
**Microsoft Account
Security Team**

02

Subject: "**Microsoft
account unusual sign
in activity**"

03

Created **urgency** by
reporting login from
Russia

04

Included **device
details** (Windows,
Chrome) to increase
credibility

05

Goal: trick users into
**securing their
account** through a
phishing link

Email Artifacts

Microsoft
account team

Display Name

The visible sender name intended to build trust with the recipient.

no-reply@access-
accsecurity.com

From Address

The actual sending address which does not belong to Microsoft.

solutionteamrecognizd0

3@gmail.co

m

Reply-To

A Gmail address used for potential direct engagement or harvesting

103.225.77.
255

Referenced IP

IP address mentioned within the content of the message body.

Authentication

Header & Authentication Analysis

```
Return-Path: <no-reply@access-accsecurity.com>
Received: from DB8EJRO6HT123.eop-KUN06.prod.protection.outlook.com (unknown [52.101.12.5])
    by mail.pot (Postfix) with ESMTP id 9F2C519A8C
    for <employee@medisurhealthnetwork.com>; Wed, 16 Aug 2023 00:15:12 +0100 (BST)
Received: from MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com (unknown [40.92.22.88])
    by DB8EJRO6HT123.eop-EUR06.prod.protection.outlook.com with ESMTP id 7DF3A27D00
    for <employee@medisurhealthnetwork.com>; Wed, 16 Aug 2023 01:14:38 +0200 (CEST)
Received: from mail.access-accsecurity.com (mail.access-accsecurity.com [89.144.44.41])
    by MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com with ESMTP id 5u3vC1k2
    for <employee@medisurhealthnetwork.com>; Wed, 16 Aug 2023 00:12:22 +0100 (BST)
Received: from atujpdtgher.co.uk (unknown [89.144.44.41])
    by mail.access-accsecurity.com (Postfix) with ESMTPA id 1C2E2350
    for <employee@medisurhealthnetwork.com>; Wed, 16 Aug 2023 00:11:12 +0200 (CEST)
Authentication-Results: mail.pot;
    spf=Fail (mail.pot: domain of access-accsecurity.com does not designate 89.144.44.41 as permitted sender);
    smtp.mailfrom=no-reply@access-accsecurity.com;
    dkim=none (no signature);
    dmarc=permanently (no valid record) header.from=access-accsecurity.com;
Received-SPF: Fail (mail.pot: domain of access-accsecurity.com does not designate 89.144.44.41 as permitted sender);
client-ip=89.144.44.41; envelope-from=no-reply@access-accsecurity.com; helo=mail.access-accsecurity.com;
X-MS-Exchange-Organization-AuthAs: Anonymous
X-MS-Exchange-Organization-AuthSource: MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com
Message-ID: <06550bd4-cd4e-4e8c-ae29-18536085bc99@mw2nam04ft048.eop-NAM04.prod.protection.outlook.com>
From: "Microsoft account team" <no-reply@access-accsecurity.com>
Reply-To: Solution Team <solutionteamrecognizd03@gmail.com>
To: employee@medisurhealthnetwork.com
Subject: Microsoft account unusual sign in activity
Date: Wed, 16 Aug 2023 00:15:00 +0100
MIME-Version: 1.0
Content-Type: text/html; charset=UTF-8
Content-Transfer-Encoding: base64
```

SPF



Failed

Sending IP was not authorized by the domain's SPF record.

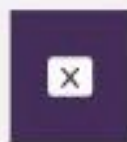
DKIM



None

There was no digital signature present in the email headers.

DMARC



Error

The policy was either misconfigured or encountered a configuration error.

Originating Infrastructure

01

Originating IP:
89.144.44.41

02

Geolocation: **Frankfurt
am Main, Germany**

03

Initial sending server:
atujpdfghher.co.uk

04

Relay server:
**mail.access-
accsecurity.com**

05

This infrastructure is
not associated with
Microsoft email
systems.

Reconnaissance Indicators

01

A **hidden tracking pixel** was detected in the email body.

02

Tracking domain:
thebandalisty.com

03

Tracking pixels allow attackers to collect **IP address and device info**.

04

Records **geographic location** and exact time email was opened.

05

This indicates **attacker reconnaissance** to identify active targets.

Log Analysis (Splunk)



Logs reviewed

Email gateway logs Proxy
browsing logs Firewall traffic
logs Authentication logs



Findings

Two users clicked the phishing
link One account likely submitted
credentials A foreign login
attempt was blocked No lateral
movement or exfiltration
observed

Risk Level

Risk Assessment

01

The incident was classified as **High Risk**.

02

Potential for **unauthorized mailbox access** and internal attacks.

03

Risk of **credential reuse** and access to sensitive communications.

04

Potential **patient data exposure** and regulatory penalties.

05

Healthcare organizations are frequent targets due to data value.

Recommendations & Conclusion



Recommendations

Enforce Multi-Factor Authentication (MFA) Strengthen DMARC email policies Improve phishing awareness training Conduct regular simulation exercises



Conclusion

Confirmed credential-harvesting attempt Security controls prevented compromise No patient data exposure identified